🔑 **Answer Key — Teacher Copy · Lesson 6-5**

MSTAR PRACTICE WORKSHEET

Lesson 6-5: Write and Evaluate Algebraic Expressions

Grade 6 Mathematics · Standard 6.AT.6a

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Which expression represents 'the product of 6 and a number n'?

- **A** 6n
- **B** 6 + n
- **C** n – 6
- **D** n ÷ 6

'Product' means multiplication, so the product of 6 and n is 6n.

Question 2 SELECTED RESPONSE — CORRECT: A

Which expression represents '9 less than a number y'?

- **A** y – 9
- **B** 9 – y
- **C** y + 9
- **D** 9y

'Less than' means subtract from the number, so it's y – 9.

Question 3 SELECTED RESPONSE — CORRECT: A

Which expression represents 'the sum of a number m and 15'?

- (A) m + 15
- (B) m – 15
- (C) 15m
- (D) m ÷ 15

'Sum' means addition, so the sum of m and 15 is m + 15.

Question 4 SELECTED RESPONSE — CORRECT: B

Evaluate 4.25s + 23.50 when s = 10.

- (A) $47.75
- (B) $66.00
- (C) $66.75
- (D) $277.50

Substitute s = 10: 4.25 × 10 + 23.50. Multiply first: 42.50, then add: 42.50 + 23.50 = $66.00.

Question 5 SELECTED RESPONSE — CORRECT: A

Which key word tells you to multiply?

- (A) Product
- (B) Sum
- (C) Difference
- (D) Quotient

'Product' means multiplication. 'Sum' means addition, 'difference' means subtraction, and 'quotient' means division.

Question 6 SELECTED RESPONSE — CORRECT: C

The expression 8n + 15 gives the cost of n tickets plus a fee. What is the cost for 6 tickets?

A $29

B $48

C $63

D $138

Substitute n = 6: 8 × 6 + 15 = 48 + 15 = $63.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: A

A concert ticket costs 8 dollars. The website also charges one 5 dollar service fee for the whole order. Let t represent the number of tickets bought in one order. Which expression gives the total cost in dollars?

(A) 8t + 5

(B) 8 + 5t

(C) 8(t + 5)

(D) 13t

The variable t is the number of tickets. Each ticket costs 8 dollars, so the ticket cost is 8t. The 5 dollar fee is charged once, so it is a constant added on: 8t + 5.

- **B:** This charges the fee on every ticket and the ticket price only once. The 8 dollars is per ticket, so it pairs with t.
- **C:** This multiplies the fee by the ticket price, charging 8 dollars on the fee as well. The fee is added once.
- **D:** This charges 13 dollars per ticket, applying the one-time fee to every ticket.

Part B — correct answer: B

Which reasoning shows why your answer to Part A is correct?

(A) The 5 dollar fee is paid for each ticket, so 5 is the coefficient: 8 + 5t.

(B) The 8 is the coefficient because it is paid for each ticket, and the 5 is a constant because it is paid one time. This gives 8t + 5, so an order of 6 tickets costs 8 × 6 + 5 = 53 dollars.

(C) Add the fee to the number of tickets first, then multiply by the ticket price: 8(t + 5).

(D) Combine 8 and 5 into a single coefficient because both are numbers: 13t.

A per-item price becomes the coefficient of the variable and a one-time charge becomes a constant. Substituting t = 6 into 8t + 5 gives 53 dollars.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, D

Select ALL statements that are true about the algebraic expression 7n + 4.

- **A** The variable is n.
- **B** The coefficient of the variable term is 7.
- **C** The constant is 4.
- **D** When n = 3, the expression has a value of 25.
- **E** When n = 3, the expression has a value of 33.
- **F** The term 7n means 7 plus n.

In 7n + 4 the variable is n, its coefficient is 7, and 4 is the constant. Substituting n = 3 gives 7 × 3 + 4 = 21 + 4 = 25. A value of 33 comes from combining 7 and 4 into 11 first, which is not allowed because 7n and 4 are not like terms. The term 7n means 7 times n.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Field day needs one $4.25 item per student plus a one-time $23.50 shipping fee. Write the expression, then find the cost for 30 students AND for 60 students. Does doubling the students double the cost? Explain.

Model answer: The expression is 4.25s + 23.50. For s = 30: 4.25 × 30 + 23.50 = 127.50 + 23.50 = $151.00. For s = 60: 4.25 × 60 + 23.50 = 255.00 + 23.50 = $278.50. Doubling the students does NOT double the cost. Twice $151.00 would be $302.00, but the actual cost is $278.50 — a difference of exactly $23.50. The reason is that only the 4.25s term depends on the number of students; the shipping fee is a constant charged once no matter how many students come, so it is not doubled along with everything else.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.